

A Dither-Based Precision Enhancement Method for Low-Bit ADCs in Spaceborne Calibration Systems

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Abstract—Space-based calibration has emerged as a critical alternative to address the limitations of traditional ground-based calibration, such as restricted field of view and atmospheric interference. However, the stringent constraints on satellite payloads often preclude the use of high-bit Analog-to-Digital Converters (ADCs), while low-bit ADCs fall short in high-precision measurement scenarios. This paper proposes a precision recovery method for low-bit ADCs based on artificial dithering. By applying a dither signal with a known distribution to the input and statistically analyzing the probability distribution of the quantized outputs across different thresholds, the method extracts high-precision measurements from low-precision hardware. Theoretical analysis proves that the estimation error's standard deviation decreases as $\Delta/\sqrt{6M}$, enabling sub-ADC-resolution accuracy through averaging. Simulations with a 4-bit ADC show that averaging $M = 200$ observations reduces the error standard deviation from 0.127 V to 0.013 V, confirming the method's ability to overcome quantization limits at negligible overhead. This offers a viable high-performance solution for resource-constrained space-based calibration systems.

Index Terms—Satellite calibration, Analog-to-Digital Converters, Artificial dithering.

I. INTRODUCTION

SATELLITE calibration is essential for ensuring the accuracy, consistency, and reliability of Earth observation data [1], [2]. A primary method is external calibration, which adjusts the radar's measurements by utilizing calibration targets on the ground [3]. This process compensates for unmodeled parameters in the antenna pattern and radar equation that are challenging to measure or estimate directly. However, a significant limitation of this ground-based approach is its sensitivity to weather, terrain, and climate variations, which constrains its accuracy and operational flexibility. In recent

years, space-based calibration has emerged as an advanced technique to overcome the constraints of conventional ground-based methods, particularly the geographical discontinuities between calibration sites and the actual observation zones [4].

Currently, the calibration of space-based equipment heavily relies on existing networks of ground-based active calibration stations [5], [6]. The core of this approach involves the use of active radiometric calibrators with precisely known radiation characteristics [7], [8]. These calibrators operate by receiving and measuring the downlink signals from satellite sensors, and subsequently transmitting high-precision calibration signals. This two-way process enables the independent and accurate calibration of both the transmit and receive chains of the sensor. Calibration is a high-precision metrological activity by its nature [9], [10]. In the resource-abundant terrestrial environment, state-of-the-art active calibration systems therefore predominantly employ high-bit and high-linearity ADCs as their core components [11], [12]. This design aims to achieve the distortion-free quantization of complex radar signals.

Compared to ground-based calibration, which is susceptible to terrain and atmospheric disturbances, space-based calibration offers distinct advantages: global coverage, immunity from atmospheric interference, and higher signal-to-noise ratios. However, the stringent payload constraints of satellite platforms necessitate extreme miniaturization and low-power design for calibration systems. This fundamental limitation creates a core contradiction: the high-precision requirements of space-based calibration are at odds with the hardware reality. Traditional high-bit ADCs, essential for high-precision measurement, are often infeasible due to their size and power consumption, while the inherent quantization inaccuracy of low-bit ADCs is insufficient to meet the demanding accuracy standards.

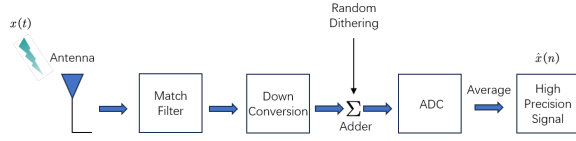


Fig. 1. System architecture.

In this paper, we propose a dither-based precision recovery method for low-bit ADCs. When an ADC measures the amplitude of a received signal, it cannot capture the exact continuous value but quantizes it to the nearest discrete level. This intrinsic quantization process introduces non-negligible errors, particularly pronounced in low-bit ADCs, fundamentally limiting the measurement accuracy. To overcome this hardware limitation, a precisely known, uniformly distributed dither signal is injected prior to quantization. Through multiple independent observations, the frequency of the quantized outputs occurring at different discrete levels is statistically recorded and analyzed. A statistical mean is then computed from these multiple quantized results. This approach enables the extraction of a high-precision amplitude estimate from the coarse outputs of a low-bit ADC. The estimation error converges as the number of observations increases, thereby achieving a measurement accuracy that surpasses the inherent quantization resolution of the ADC itself.

II. SYSTEM MODEL

As shown in Fig.1, the system architecture of the proposed dither-assisted precision enhancement scheme is illustrated in the following block diagram. The input $x(t)$ is summed with a uniform dither $d(t)$ at the adder, then quantized by a low-bit ADC $Q(\cdot)$ to produce $\hat{x}_k[n]$. After M observations, the averaging block $\frac{1}{M} \sum$ outputs the final estimate $\hat{x}[n]$. Here $x(t)$ is the continuous-time analog input signal (*e.g.*, the radar echo or calibration reference signal in a spaceborne scenario). The term $d(t)$ is the artificially injected dither signal. It is a zero-mean analog noise with a known statistical distribution, and $d(t)$ is independent of $x(t)$. The term $\hat{x}[n]$ denote the estimated high precision signals. Without the random dithering, the output of n -th sampling follows

$$\hat{x}[n] = Q(x[n]) \quad (1)$$

where $Q(\cdot)$ denotes a uniform mid-tread quantizer with a step size of Δ , which corresponds to the value of 1 Least Significant Bit (LSB). Due to the bias of low precision quantizer, we could have

$$x[n] = \hat{x}[n] + \delta[n]. \quad (2)$$

Here $\delta[n]$ is the bias leaded by quantizer. To alter the quantization behavior, a dither signal with a known statistical distribution is injected. The optimal choice is a zero-mean uniform distribution bounded to one quantization step:

$$d(t) \sim \mathcal{U}(-\Delta/2, \Delta/2). \quad (3)$$

With the artificial dithering, we could greatly improve the estimation accuracy.

III. PROPOSED METHOD

By adding artificial dithering to the analog before ADC, the output of k -th observation can be written as:

$$\hat{x}_k[n] = Q(x[n] + d_k[n]) = Q(y[n] + \delta[n] + d_k[n]). \quad (4)$$

As $y[n]$ corresponds to the discrete level, therefore $\hat{x}_k[n]$ can be rewritten as:

$$\hat{x}_k[n] = y[n] + Q(\delta[n] + d_k[n]), \quad (5)$$

It can be observed that the quantized values at the receiver will exhibit a specific distribution related to the superimposed perturbation. We achieve the multiple average and the resulting mean will be:

$$\hat{\hat{x}}[n] = \frac{1}{M} \sum_{k=1}^M \hat{x}_k[n]. \quad (6)$$

Proposition 1. Assume the artificial dithering follows a uniform distribution, the condition mean can be describable as:

$$E\{\hat{x}[n]|x[n]\} = x[n], \quad (7)$$

which indicates the conditional expectation of the quantizer output equals the true analog input.

Proof: The composite signal presented to the quantizer is $x[n] + d_k[n]$. Given the dither's limited range, the quantizer output $\hat{x}_k[n]$ can only take one of two adjacent levels:

$$\hat{x}_k[n] \in \{L^{(i)}, L^{(i+1)}\}. \quad (8)$$

The probability of $\hat{x}[n]$ equaling each level could be analyzed as follows.

For probability of $\hat{x}_k[n] = L^{(i)}$, this occurs when $x[n] + d_k[n] < (L^{(i)} + L^{(i+1)})/2 = (i + 0.5)\Delta$.

$$\begin{aligned} P(\hat{x}_k[n] = L^{(i)}|x[n]) &= P(x[n] + d_k[n] < (i + 0.5)\Delta) \\ &= P(i\Delta + \delta[n] + d_k[n] < (i + 0.5)\Delta) \\ &= P((i + \hat{\delta}[n])\Delta + d_k[n] < (i + 0.5)\Delta) \\ &= P(d_k[n] < (0.5 - \hat{\delta}[n])\Delta), \end{aligned} \quad (9)$$

where $\hat{\delta}[n] = \delta[n]/\Delta$ is normalized bias, representing the relative position of $\hat{x}_k[n]$ between $L^{(i)}$ and $L^{(i+1)}$. Given the uniform distribution of d , this probability is:

$$P(\hat{x}_k[n] = L^{(i)}|x[n]) = \frac{(0.5 - \hat{\delta}[n])\Delta - (-\Delta/2)}{\Delta} = 1 - \hat{\delta}[n]. \quad (10)$$

For probability of $\hat{x}_k[n] = L^{(i+1)}$:

$$P(\hat{x}_k[n] = L^{(i+1)}|x[n]) = 1 - P(y = L^{(i)}|x[n]) = \hat{\delta}[n]. \quad (11)$$

The condition expectation $E\{\hat{x}[n]|x[n]\}$ can be calculated as:

$$\begin{aligned}
E[\hat{x}[n]|x[n]] &= L^{(i)} \cdot P(\hat{x}_k[n] = L^{(i)}|x[n]) + L^{(i+1)} \cdot P(y = L^{(i+1)}|x[n]) \\
&= (i\Delta) \cdot (1 - \hat{\delta}[n]) + ((i+1)\Delta) \cdot \hat{\delta}[n] \\
&= i\Delta(1 - \hat{\delta}[n]) + (i+1)\Delta\hat{\delta}[n] \\
&= i\Delta - i\hat{\delta}[n]\Delta + i\hat{\delta}[n]\Delta + \hat{\delta}[n]\Delta \\
&= i\Delta + \hat{\delta}[n]\Delta \\
&= (i + \hat{\delta}[n])\Delta \\
&= x[n].
\end{aligned} \tag{12}$$

The equation $E\{\hat{x}[n]|x[n]\} = x[n]$ holds rigorously. This proves that the expected value of the coarse quantizer output is an unbiased estimate of the original fine-resolution analog input under the injection of uniformly distributed dither.

The unbiasedness property stated in Proposition 1 is not new and can be traced back to classical results on dithered quantization (e.g., [14], [15]). We include it here for self-containedness and to facilitate the subsequent variance analysis. The main novelty of this paper lies not in this property itself, but in its application to spaceborne calibration systems and the explicit derivation of the variance decay rate $\Delta^2/(6M)$ under the specific constraint of low-bit ADCs, which, to the best of our knowledge, has not been quantified in prior spaceborne calibration literature.

In practical application, we perform multiple independent observations. With the law of large numbers, this sample mean converges in probability to the population expectation:

$$\hat{\hat{x}}[n] \rightarrow E\{\hat{x}[n]|x[n]\} = x[n]. \tag{13}$$

The precision of the estimator $\hat{x}$ is quantified by its Mean Square Error (MSE), which, for an unbiased estimator, is equal to its variance $\text{Var}(\hat{x})$.

Proposition 2. Under the injection of the optimal uniform dither $d \sim \mathcal{U}(-\Delta/2, \Delta/2)$, the variance of a single dithered quantization output $\hat{x}_k[n]$ conditioned on the input $x[n]$ is given by

$$\text{Var}(\hat{x}_k[n] | x[n]) = \frac{\Delta^2}{6}. \tag{14}$$

Furthermore, if M independent observations $\{\hat{x}_k[n]\}_{k=1}^M$ are obtained, the variance of the sample mean $\hat{\hat{x}}[n] = \frac{1}{M} \sum_{k=1}^M \hat{x}_k[n]$ satisfies

$$\text{Var}(\hat{\hat{x}}[n]) = \frac{\Delta^2}{6M}. \tag{15}$$

Proof: Under the uniform dither $d \sim \mathcal{U}(-\Delta/2, \Delta/2)$ and given the input x , the quantizer output can be expressed as

$$\hat{x}_k = x + d + e_q, \tag{16}$$

where e_q is the quantization error. For a uniform mid-tread quantizer with step Δ , and because the dither amplitude exactly spans one quantization step, it is well known that the joint effect of dither and quantization results in a total error

$d + e_q$ that is independent of x and uniformly distributed over $[-\Delta/2, \Delta/2]$ [14], [15]. Consequently,

$$\text{Var}(\hat{x}_k | x) = \text{Var}(d + e_q) = \frac{\Delta^2}{12} + \frac{\Delta^2}{12} = \frac{\Delta^2}{6}, \tag{17}$$

where the two $\Delta^2/12$ terms come from the variance of the uniform dither and the uniform quantization error, respectively. The first equality follows from the independence of d and e_q under the given conditions.

Since the M observations are i.i.d., the variance of the sample mean is

$$\text{Var}(\hat{\hat{x}}) = \frac{1}{M} \text{Var}(\hat{x}_k | x) = \frac{\Delta^2}{6M}. \tag{18}$$

The advantages of this method can be summarized as follows:

- 1) **Attainable Precision:** The standard deviation of the estimate, $\sqrt{\text{Var}(\hat{\hat{x}})} = \Delta/\sqrt{6M}$, represents the attainable measurement precision. It can be made significantly smaller than the ADC's intrinsic resolution Δ by increasing M .
- 2) **Effective Number of Bits (ENOB):** The method effectively increases the system's resolution. The effective quantization step is reduced to $\Delta_{\text{eff}} = \Delta/\sqrt{M}$. This translates to an increase in the ENOB by approximately $\frac{1}{2} \log_2 M$ bits compared to the native ADC.
- 3) **Convergence Rate:** The estimation error decreases with the square root of the number of observations ($1/\sqrt{M}$). To double the precision, one needs to quadruple the number of observations M .

IV. SIMULATION

To verify the theoretical analysis presented in the previous sections and to illustrate the effectiveness of the dithering and averaging method in enhancing ADC resolution, a comprehensive simulation study was conducted. This section details the setup, parameters, and objectives of the simulation experiments. In each simulation trial, a deterministic sinusoidal signal with a fixed amplitude and frequency serves as the fine-resolution analog input. The signal is first added with a uniformly distributed dither $d[t] \sim \mathcal{U}(-\Delta/2, \Delta/2)$, and the sum is then quantized by a uniform mid-tread quantizer to generate the low-bit digital output.

For clarity, a baseline scenario is configured with a 4-bit analog-to-digital converter (ADC) with a full-scale range of $[0, 5]$ V, yielding a quantization step size of $\Delta = 5/2^4 = 0.3125$ V. The input signal is a composite sinusoid with fundamental frequency $f_0 = 20$ Hz, sampled at $f_s = 1$ kHz over a duration of $T = 0.05$ s, resulting in $N = 50$ samples per realization. Prior to quantization, independent uniform dither $d \sim \mathcal{U}(-w, w)$ with amplitude $w = 0.15$ V is added to each sample.

Fig. 2 illustrates the effect of averaging on 4-bit ADC quantization with uniform dithering. The original multi-tone signal (blue curve) is corrupted by quantization noise when sampled

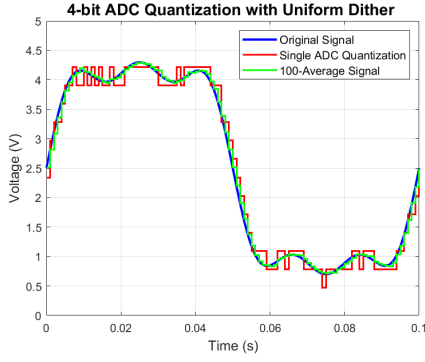


Fig. 2. The 4-bit ADC quantization with uniform dithering.

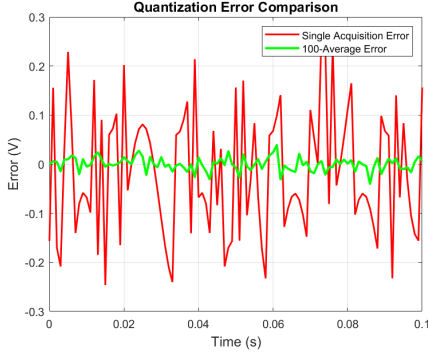


Fig. 3. Quantization errors for single versus averaged acquisitions.

once (red staircase), exhibiting characteristic step-like distortions. After averaging 100 independent dithered-quantized observations, the reconstructed signal (green staircase) shows significantly reduced quantization artifacts while maintaining the fundamental signal shape. The quantization levels are marked by gray dashed lines, revealing how averaging allows the system to effectively resolve intermediate voltages between discrete ADC levels, thereby improving amplitude resolution beyond the nominal 4-bit limitation.

Fig. 3 compares quantization errors for single versus averaged acquisitions. The single-acquisition error (red) fluctuates with peak amplitudes approaching ± 0.156 V, corresponding to the theoretical quantization step boundary (black dashed lines). In contrast, the 100-average error (green) demonstrates substantial reduction in both amplitude and variability, with errors tightly clustered near zero. This temporal error reduction directly translates to the observed variance improvements, confirming that multiple independent observations effectively cancel out uncorrelated noise components while preserving the underlying signal information.

Fig. 4 shows the error standard deviation versus averaging count M . Experimental results (red) closely follow the theoretical $1/\sqrt{M}$ decay (blue dashed line), decreasing from 0.1269 V ($M = 1$) to 0.0128 V ($M = 200$). This 14-fold reduction aligns well with the theoretical $\sqrt{M}$ improvement factor, validating that averaging effectively suppresses random errors in dithered quantization systems. Minor deviations at high M are attributed to ADC resolution limits.

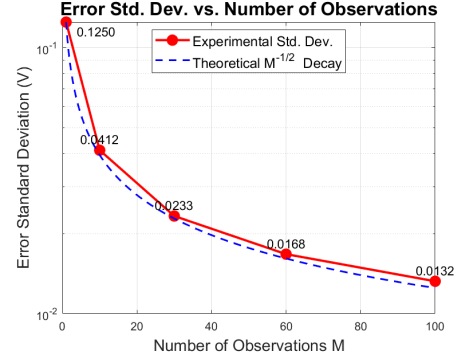


Fig. 4. Error standard deviation versus averaging count M .

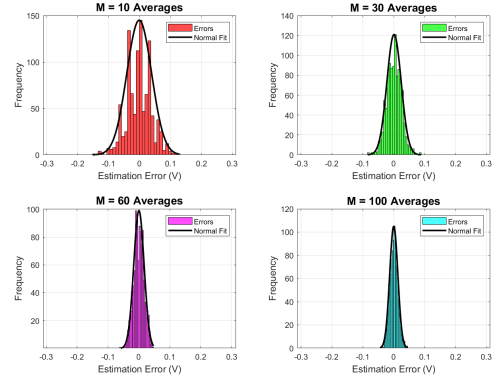


Fig. 5. Error distributions.

Fig. 5 shows error distributions for $M = 10, 30, 60, 100$ averages. As M increases, the histograms become narrower and more Gaussian-shaped (black fitted curves), confirming the Central Limit Theorem's prediction. The distributions shift from asymmetric (due to quantization artifacts) toward symmetric Gaussian forms, with standard deviations decreasing proportionally to $1/\sqrt{M}$. This demonstrates that averaging not only reduces error magnitude but also makes the error statistics more predictable.

V. CONCLUSION

This paper introduces a dither-assisted precision enhancement method for low-bit ADCs in space-based calibration systems. By injecting a known dither signal and averaging multiple quantized observations, the system achieves high-resolution signal recovery beyond the ADC's intrinsic limits. Theoretical analysis demonstrates that the estimation error decreases with the number of observations, enabling effective resolution improvement without additional hardware complexity, and simulation results confirm the theory analysis. It is worth noting that, although the central limit theorem ensures asymptotic Gaussianity, the error distributions for small to moderate M (e.g., $M = 10, 50$ in Fig. 5) exhibit residual multimodality due to the discrete nature of quantization. Addressing this limitation via finite-sample corrections or alternative dither designs is left for future work.

REFERENCES

- [1] C. Lu, T. Yang, B. Zhang and H. Yu, "Spaceborne Altimeter Calibration System With Calibration Satellites," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 17, pp. 18613-18621, Sep. 2024.
- [2] M. Shimada, O. Isoguchi, T. Tadono, and K. Isono, "PALSAR radiometric and geometric calibration," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 47, no. 12, pp. 3915–3932, Dec. 2009.
- [3] L. B. Kunz and D. G. Long, "Calibrating SeaWinds and QuikSCAT scatterometers using natural land targets," in *IEEE Geoscience and Remote Sensing Letters*, vol. 2, no. 2, pp. 182-186, Apr. 2005.
- [4] T. Qiu, Y. Wang, J. Hong, K. Xing, S. Du, and J. Mu, "An in-orbit measurement method for elevation antenna pattern of meo synthetic aperture radar based on nano calibration satellite," in *Remote Sensing*, vol. 14, no. 3, Feb. 2022.
- [5] G. Liu, L. Li, J. Hong and F. Ming, "Shift-Frequency Jamming Imaging and Analysis Based on Active Radar Calibrator," in *2021 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Brussels, Belgium, 2021, pp. 3948-3951.
- [6] C. Tang, T. Ni, Y. Bai, X. Li, C. Sun, and C. Li, "A novel phase bias calibration algorithm for multi-channel HRWS SAR system in azimuth," in *2023 4th China International SAR Symposium (CISS)*, 2023, pp. 1–4.
- [7] F. Zhang, X. Yao, H. Tang, Q. Yin, Y. Hu and B. Lei, "Multiple Mode SAR Raw Data Simulation and Parallel Acceleration for Gaofen-3 Mission," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 11, no. 6, pp. 2115-2126, Jun. 2018.
- [8] M. Shimada, H. Oaku, and M. Nakai, "SAR calibration using frequency-tunable active radar calibrators," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 37, no. 1, pp. 564–573, Jan. 1999.
- [9] X. -Y. Xu, K. Xu, H. Shen, Y. -L. Liu and H. -G. Liu, "Sea Surface Height and Significant Wave Height Calibration Methodology by a GNSS Buoy Campaign for HY-2A Altimeter," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 9, no. 11, pp. 5252-5261, Nov. 2016.
- [10] X. Wang, L. Liu, H. Shi, X. Dong and D. Zhu, "In-orbit calibration and performance evaluation of HY-2 scatterometer," in *2012 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Munich, Germany, 2012, pp. 4614-4616.
- [11] C. Simmonds and M. Beach, "Downlink calibration requirements for the TSUNAMI (II) adaptive antenna testbed," in *9th IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, vol. 3, 1998, pp. 1260 – 1264.
- [12] X. Huang, V. Dyadyuk, Y. J. Guo, L. Stokes, and J. Pathikulangara, "Frequency-domain digital calibration and beamforming with wideband antenna array," in *2010 IEEE Global Telecommunications Conference (GLOBECOM)*, 2010, pp. 1–5.
- [13] J. Liu, X. Tang, F. Wang, X. Chen and W. Xu, "A High-Resolution Channel Estimation Method for ICAN Systems Based on Artificial Perturbation-OTFS," in *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 4561-4576, Dec. 2026.
- [14] R. M. Gray and T. G. Stockham, "Dithered quantizers," *IEEE Transactions on Information Theory*, vol. 39, no. 3, pp. 805–812, May 1993.
- [15] J. Chen, Y. Wang, and M. K. Ng, "Quantized low-rank multivariate regression with random dithering," *IEEE Transactions on Signal Processing*, Oct. 2023.